

SUSHANT KUMAR

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EDUCATION

Indian Institute of Technology, Delhi

2022 to 2026 (Expected)

B.Tech in Electrical Engineering | Dept. change Mech. to Elec. Engg., 2023

- Relevant coursework: Data Structures & Algorithms, Computer Architecture, Machine Intelligence & Learning, Discrete Mathematics, Optimisation Methods, Probability & Stochastic Processes.

SOFTWARE ENGINEERING PROJECTS

Real-Time Social Feed Platform

TypeScript, Node.js, Express, Redis, PostgreSQL, React, Docker, NGINX, k6

- Real-time collaborative feed: posts, threaded replies, reactions, live presence and typing indicators. Solved cross-instance fan-out with a WebSocket layer backed by Redis Pub/Sub; presence tracked via heartbeats and TTL expiry, so disconnected users are evicted rather than lingering as ghosts.
- Replaced offset pagination after duplicate and missing posts showed up under concurrent writes: opaque cursors encoding timestamp plus post ID restored stable ordering. Ranking pass blends recency, engagement, and author affinity.
- Load tested with k6. First run capped near 3,500 sockets per instance on file descriptor limits; after kernel and FD tuning plus a horizontally scaled socket tier behind a sticky-session balancer, sustained 25,000 concurrent connections at p50 18 ms, p95 42 ms, p99 85 ms.

Experimentation and Feature Flag Platform

Python, FastAPI, Kafka, PostgreSQL, Redis, React, TypeScript, SciPy, Docker

- Flag evaluation, experiment assignment, exposure logging, metric aggregation, and analysis. Assignment is deterministic (user ID hashed with a per-experiment salt), so no central assignment service is needed and experiments stay independent.
- Client SDK caches configs and evaluates flags in-process with stale-while-revalidate refresh, keeping rendering off the network path and apps responsive during backend outages.
- Exposure events flow through Kafka into aggregated metrics: two-sample t-tests on primary metrics, guardrails on latency, crash, and error rates. A Monte Carlo study (100,000 synthetic users, 10,000 runs) showed naive daily peeking inflated false positives from 5% to roughly 14%, so I added a sequential-testing correction for valid continuous monitoring.
- Chi-square sample ratio mismatch detection alerts at $p < 0.01$, catching broken experiments before decisions are made off invalid data. Dashboard shows confidence intervals, guardrails, and a one-click kill switch.

AI Code Review Agent with Evaluation Infrastructure

Python, FastAPI, LangChain, PostgreSQL, GitHub Actions, Docker, Prometheus, Grafana, PyTest

- Agent reviews pull requests for correctness, maintainability, style, documentation, and test coverage; most of the work was the evaluation and release harness that makes prompt and model changes measurable and safe to ship.
- Curated and labelled a 1,200-example golden set covering common cases, edge cases, and adversarial inputs. Automated eval pairs deterministic checks (schema, latency, cost) with LLM-as-judge scoring calibrated to human labels.
- Wired the suite into CI/CD as a regression gate: any prompt, retrieval, or model change that drops aggregate scores below threshold fails the build, with prompt versions stored alongside source for reproducible comparison. Orchestration handles structured output validation, repair of malformed responses, exponential-backoff retries, and fallback routing on timeout or rate limit.
- Cost and latency telemetry showed that routing style-only reviews to a smaller model cut inference cost 68% at equal quality, near \$1.80 per 1,000 requests.
- PII redaction before any external model call, confidence-based escalation of uncertain reviews to humans, bias auditing across Python, Java, JavaScript, and C++, and full logging of inputs, outputs, confidence, and latency. In an internal GitHub workflow it cut average review turnaround 52%.

RESEARCH AND INDUSTRY EXPERIENCE

Research Contributor, RTLGen Project (Remote)

Jan to Jul 2024

Edge Computing Lab, Harvard University | Prof. Vijay Janapa Reddi. Contributed to the RTLGen Bench Harness architecture; built LangChain pipelines for automated prompt-feedback evaluation of generated hardware designs. Acknowledged in the resulting publication (arXiv:2501.01892).

Computer Vision Intern, Autonomous Drone Navigation

Jul to Nov 2024

Botlab Dynamics Pvt. Ltd., Unmanned Aerial Systems Division. Built real-time object tracking and depth-based obstacle awareness for UAV navigation on resource-constrained edge hardware; analysed flight-test telemetry to diagnose detection failures.

Research Intern to B.Tech Project, Clinical Video Analytics

May 2025 to present

Neuro Computing Lab, IIT Delhi | Prof. Tapan Kumar Gandhi & Dr. Manvir Bhatia. Detection pipeline for sub-pixel involuntary movement in long-duration patient video (de-noised optical flow, adaptive temporal smoothing, dynamic thresholding): 40%+ improvement over baseline, validated against clinician-annotated ground truth.

Co-author, Hyperbolic Geometry of Reasoning in LLM Hidden States

2025

Submitted to Mechanistic Interpretability Workshop @ ICML 2026. Poincaré-ball probe over the hidden states of a 7B model recovered reasoning depth at Spearman $\rho = 0.967$ against $\rho = 0.488$ for the matched Euclidean baseline.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, Java, MATLAB, SQL (basic), Verilog.

Backend and infrastructure: Node.js, Express, FastAPI, WebSockets, Redis, PostgreSQL, Kafka, NGINX, Docker, GitHub Actions, Prometheus, Grafana, k6; AWS and GCP (coursework). **Testing and analysis:** PyTest, A/B testing, sequential testing, hypothesis testing, pandas, SciPy, NumPy.

Frontend and AI tooling: React, optimistic UI, cursor-based pagination; LangChain, LLM-as-judge evaluation harnesses, prompt and context engineering, PyTorch, TensorFlow, scikit-learn.

ACHIEVEMENTS

- **Lead Organiser, Yann LeCun Panel at IIT Delhi (Oct 2024):** led a 3-person team for a 5,000-attendee event with Meta's Chief AI Scientist; primary liaison with Meta's global comms team.
- **JEE Advanced 2022:** Rank 417 (top 0.2%). Indian Mathematics Olympiad Qualifier (2020), UP region. Member, IIT Delhi Institute Football Team, BSA General Championship winners.